

System Architecture

Boundaries, services, topology, and principal request flows.

Eurskem AI · Agentic Workflow Platform

Generated 2026-08-24

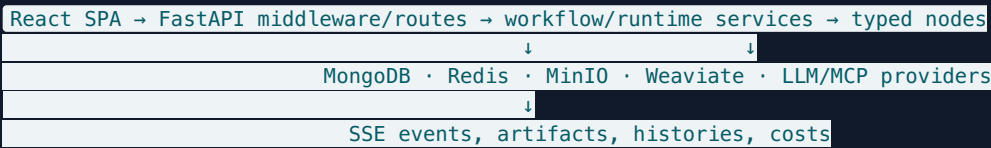
Code revision 8bd16ca4a2a3

Derived from the live repository

Source of truth. Inventories and symbol tables are generated from the current checkout. Narrative explains observed structures and does not replace executable contracts.

Architectural model

The browser communicates through HTTP and authenticated event streams. FastAPI routes validate identity and payloads, invoke runtime or store functions, and persist state through MongoDB, MinIO, Weaviate, and Redis-backed coordination. The compiler turns YAML into a LangGraph execution graph whose nodes resolve from `NodeRegistry`.



Boundary rules

- UI code calls backend APIs and never imports backend modules.
- API routes own authentication and HTTP error translation.
- Runtime owns graph semantics and delegates capability work to nodes.
- Nodes receive infrastructure through shared services.
- Persistent stores define durable user-visible state.

Package / area	Files	Symbols	Representative responsibility
app	3	18	Python module.
app/api	29	382	Python module.
app/db	3	25	Python module.
app/evaluation	6	29	Evaluation package: LLM-as-a-Judge harness.
app/evidence	6	54	Proposal-grade evidence gathering and citation verification.
app/ingestion	14	129	Python module.
app/integrations	21	186	Python module.
app/knowledge	6	90	Knowledge Studio control plane: stable resources, profiles and indexes.
app/llm	18	183	LLM gateway package.
app/mcp	28	268	Real, live MySQL-backed business-records MCP server.
app/nodes	62	603	Built-in workflow-node discovery.
app/observability	5	17	Python module.
app/proposal_graph	9	68	Proposal workspace graph: domain state and workspace store.
app/rag	3	7	Saved RAG Agent resources and grounded generation.
app/research	3	30	Controlled adapters for external scientific research assets.
app/retrieval	16	69	Hybrid RAG retrieval pipeline.
app/runtime	21	272	Phase 4 runtime: YAML to LangGraph compiler and executor.

Package / area	Files	Symbols	Representative responsibility
app/security	17	94	Python module.
app/storage	2	12	Python module.
app/tools	14	104	Character-count helper for the proposal-size guardrails.
app/workflow	16	170	Durable, file-backed state used by the Workflow Builder.

File	Responsibility
app/main.py	FastAPI application entry point for Eurskem AI — Agentic Workflow Platform.
app/config.py	Config module.
app/db/mongo.py	Async MongoDB client and manifest collection.
app/runtime/events.py	Bounded, session-scoped pub/sub for workflow Server-Sent Events.
app/runtime/coordination.py	Small Redis coordination primitives shared by Uvicorn workers.
docker-compose.yml	services: app: build: context: .
docker-compose.production.yml	name: eurskem-ai x-logging: &default-logging driver: json-file options: max-size: "10m" max-file: "5" services: caddy: image: caddy:2.10.0-alpine restart: unless-stopped ports: - "80:80" - "443:443" - "443:443/udp" environment: DOMAIN: \${DOMAIN:?Set DOMAIN in .env.production} ACME_EMAIL: \${ACME_EMAIL:?Set ACME_EMAIL in .env.production} MAX_REQUEST_BODY_SIZE: \${MAX_REQUEST_BODY_SIZE:-220MB} volumes: - ./deploy/ionos/Caddyfile:/etc/caddy/Caddyfile:ro - caddy-data:/data - caddy-config:/config depends_on: app: condition: service_healthy frontend: condition: service_healthy networks: - edge logging: *default-logging frontend: build: context: ./ui args: VITE_API_URL: "" restart: unless-stopped networks: - edge healthcheck: test: ["CMD-SHELL", "wget -q --spider http://127.0.0.1:8080/healthz"] interval: 20s timeout: 5s retries: 5 start_period: 10s read_only: true tmpfs: - /var/cache/nginx - /var/run - /tmp logging: *default-logging app: build: context: .